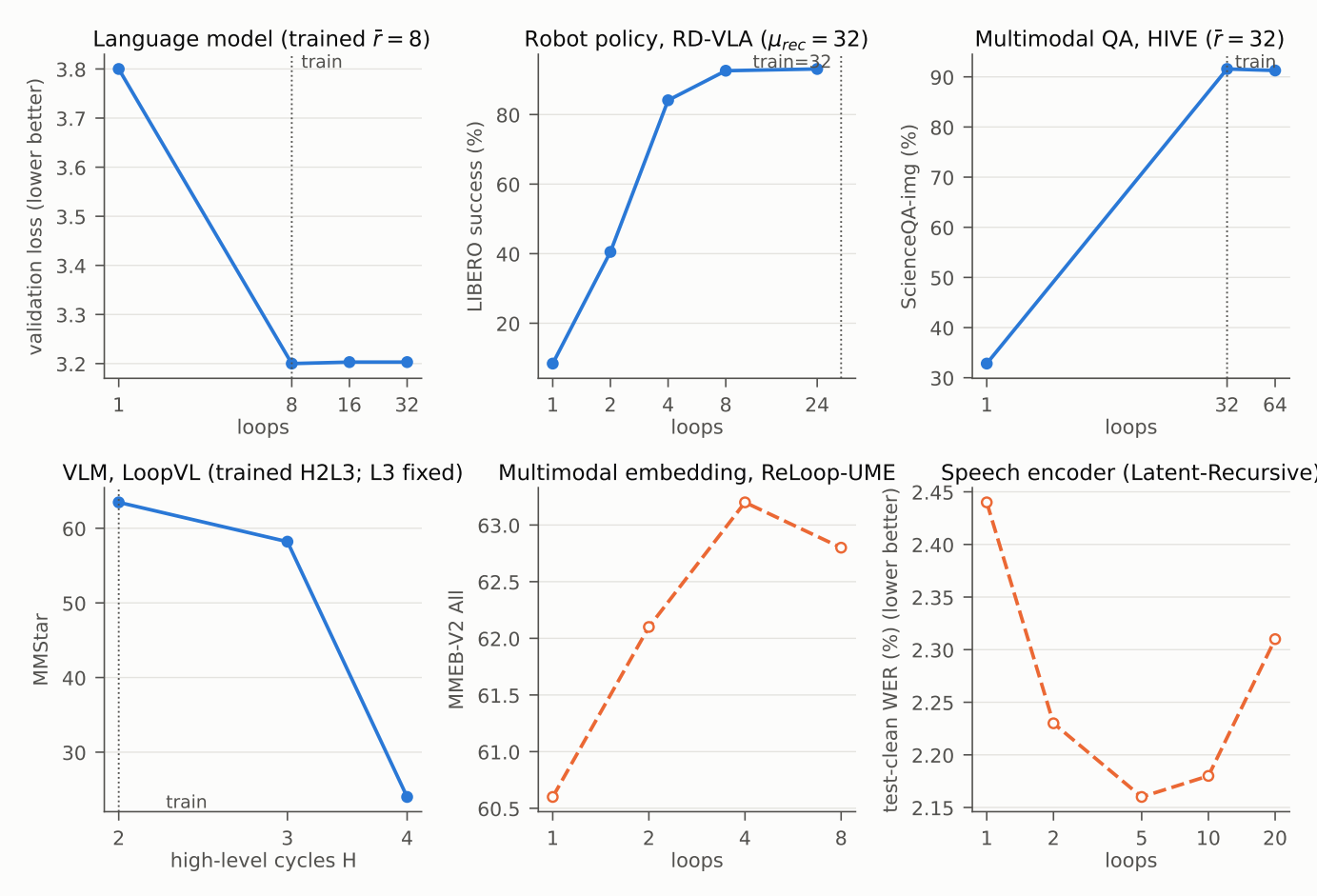
Language model (trained $\bar{r} = 8$)
validation loss (lower better)
3.8
3.7
3.6
3.5
3.4
3.3
3.2
train
1
8
16
32
loops
Robot policy, RD-VLA ($\mu_{rec} = 32$)
LIBERO success (%)
80
60
40
20
train=32
1
2
4
8
24
loops
Multimodal QA, HIVE ($\bar{r} = 32$)
ScienceQA-img (%)
90
80
70
60
50
40
30
train
1
32
64
loops
VLM, LoopVL (trained H2L3; L3 fixed)
MMStar
60
50
40
30
train
2
3
4
high-level cycles H
Multimodal embedding, ReLoop-UME
MMEB-V2 All
63.0
62.5
62.0
61.5
61.0
60.5
1
2
4
8
loops
Speech encoder (Latent-Recursive)
test-clean WER (%) (lower better)
2.45
2.40
2.35
2.30
2.25
2.20
2.15
1
2
5
10
20
loops